Week 12: Understanding and World Models in Large Language Models

Philosophy of Artificial Intelligence

Philosophy of Artificial Intelligence (11118BLG003). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

Large language models write fluent text on almost any topic. Whether they understand what they write is one of the most contested questions in AI today. This week examines the debate, the distinction between formal and functional linguistic competence, evidence for internal world models and the risks of anthropomorphic description [1, 2, 3].

Learning outcomes

By the end of the week, students are expected to summarise the arguments for and against understanding in language models. They should be able to distinguish formal from functional linguistic competence. They should also be able to explain what counts as evidence for a world model, including interventions on internal states. Finally, they should be able to apply Shanahan's cautions to descriptions of these systems.

Lecture

The debate

Mitchell and Krakauer surveyed a heated debate in the AI research community on whether large pretrained language models can be said to understand language and the situations it describes [1]. Those who answer yes point to performance on demanding tasks, to abilities that appear with scale and to internal representations that track structure in the world. Those who answer no point to the lack of grounding discussed in Week 5 and to brittleness on problems that differ slightly from familiar ones. They also note that sophisticated pattern completion can produce fluent text. Mitchell and Krakauer suggested that the debate may reflect distinct modes of understanding, and they called for an extended science of intelligence that could characterise them. Earlier, Floridi and Chiriatti had tested GPT-3 on mathematical, semantic and ethical questions and described it as a system that produces text without understanding it [4].

Formal and functional competence

Mahowald and colleagues proposed a distinction that clarifies much of the disagreement [2]. Formal linguistic competence is knowledge of the rules and statistical regularities of a language: grammar, morphology and the patterns of word use. Functional linguistic competence is the ability to use language in the world, which draws on reasoning, world knowledge, situation modelling and social cognition. In the human brain, the two rely on different networks. Language models, they argued, have achieved impressive formal competence, while their functional competence is uneven and often depends on additional training or tools. On this view, fluent language is not a reliable sign of thought, a lesson that recalls the ELIZA effect of Week 2.

World models

A world model, in the sense relevant here, is an internal representation that tracks the hidden state of a domain and supports predictions under changes, including counterfactual ones. Li and colleagues trained a sequence model only on legal move sequences of the board game Othello, without any description of the board [3]. They found that the network's internal activations encoded the state of the board. Intervening on these activations to change the represented state changed the model's predictions of legal moves accordingly. The intervention matters: A correlation between activations and board states could be incidental, but a causal role in prediction suggests that the representation is used.

The interactive lab and the Colab notebook reproduce the logic of such tests in a transparent setting. Tokens describe the moves of an agent in a small grid, and some moves bump into a wall. A predictor that uses only the last few tokens gains from a short context. It loses accuracy when longer contexts become too rare to estimate from the data. As Figure 12.1 shows, it never reaches the accuracy of a predictor that tracks the hidden position. Intervening on the tracked position changes the tracker's predictions, while the n-gram predictor has no state to intervene on.

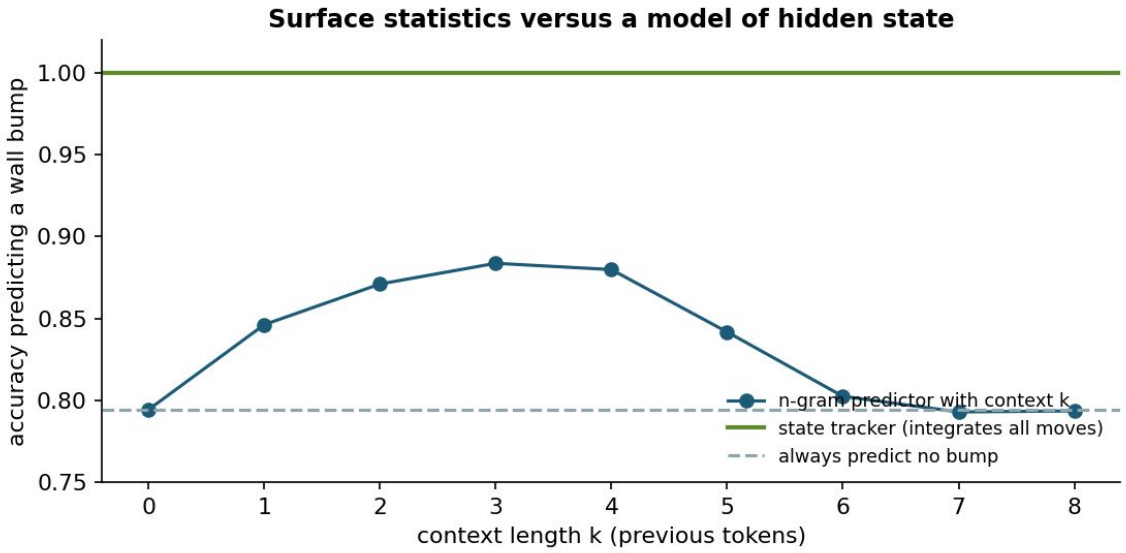


Figure 12.1. Accuracy of predicting a wall bump in a small grid world. Predictors based on n-grams gain from short contexts but lose accuracy when longer contexts become too rare to estimate. They remain below a predictor that tracks the hidden position.

Talking about language models

Shanahan warned against describing language models with words such as knows, believes and thinks without care [5]. A bare language model is a system that predicts the next token, and much of its apparent agency comes from the dialogue systems and prompts built around it. Anthropomorphic descriptions can be useful shorthand, but they may lead users to attribute beliefs and reliability that the system does not have. Bender and Koller's argument from Week 5 adds that fluency is not evidence of meaning [6]. Week 14 returns to the question of when attributing beliefs to a system is legitimate.

> ***Pause and reflect.*** *What result would convince you that a language model understands a domain you know well? Would it be a behavioural test, an internal analysis or both?*

Hands-on components

Interactive lab: An agent walks randomly in a 5 by 5 grid, and the token stream records each move and whether it bumped into a wall. Compare an n-gram predictor with a state tracker, then intervene on the tracker's internal position, in the spirit of intervention experiments on sequence models [3]. Link: https://utkukose.github.io/philosophy-of-ai-NB-lecture/weeks/week-12/lab.html

Colab notebook: It generates token streams from a hidden grid walk and fits n-gram predictors of increasing context length. It compares them with a state tracker and performs an intervention on the tracker's state [1, 3]. Link: https://colab.research.google.com/github/utkukose/philosophy-of-ai-NB-lecture/blob/main/weeks/week-12/NB12_understanding_llms.ipynb

Reflection prompts

1. Which of the two competences does your own use of language models rely on most, and where has the other one failed you?
2. Is a representation that supports interventions sufficient for understanding, or only necessary? Argue your case.
3. Rewrite a sentence from a news report about AI that uses mental vocabulary, following Shanahan's advice.

Weekly task and submission

What to do. Evaluate whether a present language model understands one domain of your choice. First, apply the distinction between formal and functional competence to that domain. Second, propose one behavioural test and one internal test, such as a probe of internal states or an intervention on them. Third, explain what each test could show and what it could not show.

Where to look. Lecture: sections "Formal and functional competence", "World models" and "Talking about language models". Interactive lab: the comparison of an n-gram predictor with a state tracker, and the intervention on the tracker's internal position (tab: Explore). Colab notebook: Section 2 for n-gram predictors and Section 3 for the state tracker and the intervention.

Example. For chess, a behavioural test could ask the model for legal moves in unusual positions. An internal test could follow the Othello study: Change the represented board state and check whether the predicted moves change accordingly.

What to submit. About 500 words, and the Colab notebook with Exercises 1 and 2 completed.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

References

[1] Mitchell, M., & Krakauer, D. C. (2023). The debate over understanding in AI's large language models. Proceedings of the National Academy of Sciences, 120(13), e2215907120. https://doi.org/10.1073/pnas.2215907120

[2] Mahowald, K., Ivanova, A. A., Blank, I. A., Kanwisher, N., Tenenbaum, J. B., & Fedorenko, E. (2024). Dissociating language and thought in large language models. Trends in Cognitive Sciences, 28(6), 517-540. https://doi.org/10.1016/j.tics.2024.01.011

[3] Li, K., Hopkins, A. K., Bau, D., Viégas, F., Pfister, H., & Wattenberg, M. (2023). Emergent world representations: Exploring a sequence model trained on a synthetic task. In International Conference on Learning Representations (ICLR). https://arxiv.org/abs/2210.13382

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[5] Shanahan, M. (2024). Talking about large language models. Communications of the ACM, 67(2), 68-79. https://doi.org/10.1145/3624724

[6] Bender, E. M., & Koller, A. (2020). Climbing towards NLU: On meaning, form, and understanding in the age of data. In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (pp. 5185-5198). https://doi.org/10.18653/v1/2020.acl-main.463